

Krishna Pratap Singh

✉ krishna191217@gmail.com | 📞 +91 7017737354 | [🌐 krishna99dev](https://github.com/krishna99dev)

Skills

C/C++, Python, SQL, JavaScript, HTML5, CSS3 | React.js, Flask, Django | MongoDB (NoSQL), Relational Databases | Cloud Platforms (Azure, AWS (EC2), GCP),| Git/GitHub, Linux | TypeScript, SOLID Design Principles, Software Design Principles, RESTful APIs, API Integration, JWT | Pandas, NumPy, Scikit-learn | Jira, CI/CD, Agile Methodologies/Scrum | Full Product Development Cycle, Debugging, Scalability, Profiling | Design Patterns, GitHub Copilot
Soft Skills: Problem-Solving, Communication, Agile Teamwork, Adaptability, Attention to Detail, Time Management, Analytical Thinking, Leadership, Cross-Functional Collaboration, Accountability, Cross-functional Collaboration.

Work Experience

BDH IT Solutions Jan 2026 – Present
Associate Software Engineer

- Built a web-based anonymous vehicle contact system in **React** and **Next.js** — scanner side loads in **<500ms** with no app or sign-up needed; owner side runs as a **PWA dashboard** to receive and manage incoming contact alerts.
- Engineered a **Next.js Edge Runtime gateway** that maps each physical QR tag to a unique backend token in sub-50ms, routing public scanners to the contact flow while keeping all owner **PII server-side**.
- Integrated **Exotel Connect API** to bridge anonymous PSTN calls between scanner and owner by **CallerId masking** (<3s low latency), backed by **Upstash Redis** for session management and **MongoDB Atlas** for tag-to-owner mapping.

VS Associates Sep 2024 – Jan 2026
Junior Software Engineer

Employee Activity Monitoring System | React.js, JWT, Redis

- Built Employee Activity Monitoring System frontend in **React.js**, delivering responsive, individual and team-level metrics dashboards for **100+** employees.
- Reduced dashboard load time by **40%** using **React.memo**, custom hooks, and component-level state isolation, optimizing render cycles and performance across 100+ employee metric views.
- Architected **Role-Based Access Control (RBAC)** and stateless **JWT** auth with **Redis**-managed refresh tokens, securing dashboard access for 100+ employees with scalable, secure session handling.

Data Visualization Application | React.js, DSU, AWS

- Led frontend development in **React.js**, delivering intuitive, accessible, and interactive user interfaces for data visualization.
- Implemented **Union-Find (DSU)** to dynamically group **1,000+** client relationship graphs into connected family clusters, enabling scalable rendering of complex relationship networks.
- Deployed on **Amazon Web Services (AWS)** cloud platform, accelerating asset delivery and reducing initial load times.

Panda Tech Consultancy March 2024 – Aug 2024
Trainee Software Developer

- Built reusable **React.js** components with the business team, accelerating frontend delivery across multiple screens.
- Integrated **RESTful APIs** into UI workflows, used **Postman** for API debugging, and maintained technical documentation to support team onboarding and code reviews.

Defense Research and Development Organization (DRDO) May 2023 – June 2023
Software Internship

- Developed a **Flask**-based ML web backend service to predict startup acquisition likelihood using model classifiers.
- Engineered data preprocessing pipelines using **pandas** and **NumPy** to clean and transform **500+** startup records into features optimized for model training and evaluation.

Education

Amity University, Noida | B.Tech in Information Technology Aug 2020 – Jul 2024

Achievements

- 1st place** for ARITech, an innovative AR-based learning project, at InCITe 2023, Amity University.
- Optimizing Supply Chain of Beauty Products Using NSGA-II Algorithm** — Co-authored and published in **Springer Nature** (Nov 2025), recognized at ICDIS 2024 and other conferences in AI and smart systems.
- CCNA certification** demonstrating proficiency in foundational networking concepts.

Leadership/Extracurricular

- President, E-connect Club:** Organized tech events and workshops for 300+ students, driving hands-on learning.
- Lead Organizer & Tech Lead:** Spearheaded CyberHack, AYF (500+ participants) and engineered the official **InCITe 2023** conference website, boosting attendee engagement by **30%** through responsive design and intuitive UX.